

CCL VIVA MASTER PREPARATION

STEP 1 — ALL CONCEPTS IN SIMPLE WORDS

Cloud Computing

Instead of buying your own computer/server, you rent one over the internet and pay only for what you use. Like renting a room in a hotel instead of buying a house.

Real-life example: Gmail, Google Drive, Netflix — all run on cloud. You don't own the server, you just use it.

Why it is used: Saves money, no maintenance, access from anywhere, scales up when needed.

"Cloud computing is the delivery of computing services like servers, storage, and databases over the internet on a pay-as-you-go basis."

Virtualization

Running multiple fake computers inside one real computer. Like dividing one big hall into multiple rooms using temporary walls.

Real-life example: Your college lab has one powerful server but 30 students can each use their own "virtual" machine on it.

Why it is used: Saves hardware cost, uses resources efficiently, easy to create/delete machines.

"Virtualization is the process of creating multiple virtual machines on a single physical machine using software called a hypervisor."

Hypervisor

The software that creates and manages virtual machines. It sits between the hardware and the VMs and divides resources among them.

Real-life example: Think of a hypervisor like a hotel manager — one hotel (server), many rooms (VMs), manager (hypervisor) decides who gets which room.

Why it is used: To run multiple operating systems on one physical machine simultaneously.

"A hypervisor is software that creates and manages virtual machines by dividing physical hardware resources among them."

Type 1 vs Type 2 Hypervisor

Simple explanation:

- **Type 1 (Bare-metal):** Runs directly on the hardware. No host OS needed. Like building a hotel directly on land.
- **Type 2 (Hosted):** Runs on top of an existing OS. Like building a hotel on top of another building.

Feature	Type 1	Type 2
Runs on	Bare hardware	Host OS
Performance	High	Lower
Examples	Xen, Hyper-V	VirtualBox, KVM
Used in	Data centers	Personal computers

"Type 1 hypervisor runs directly on hardware — examples are Xen and Hyper-V. Type 2 runs on top of a host OS — examples are VirtualBox and KVM."

Virtual Machine (VM)

A software-based computer that runs inside a real computer. It has its own OS, RAM, storage — but it's all fake (virtual).

Real-life example: Running Ubuntu Linux inside your Windows laptop using VirtualBox.

Why it is used: Test software without affecting your main machine, run multiple OS, save hardware cost.

"A virtual machine is a software-based computer that runs its own operating system inside a physical computer."

Host OS vs Guest OS

- **Host OS** = The real operating system running on your actual hardware (e.g., Windows on your laptop)
- **Guest OS** = The operating system running inside the virtual machine (e.g., Ubuntu inside VirtualBox)

"Host OS runs on physical hardware; Guest OS runs inside a virtual machine on top of the host."

KVM (Kernel-based Virtual Machine)

KVM is a Type 2 hypervisor built into the Linux kernel itself. When you install KVM, your Linux computer becomes capable of running VMs with near-hardware performance.

Real-life example: Like upgrading your car's engine so it can now tow heavy trailers — Linux gets upgraded to run VMs.

Why it is used: Free, open-source, high performance, built into Linux — no extra software needed.

"KVM is a Type 2 hypervisor built into the Linux kernel that allows Linux to run virtual machines using hardware virtualization."

VirtualBox

A free software by Oracle that lets you run virtual machines on your Windows, Mac, or Linux computer. Very easy to use with a graphical interface.

Real-life example: Like installing an app on your computer — you open VirtualBox, click "New VM," and run Ubuntu inside Windows.

Why it is used: Free, easy, cross-platform — perfect for students and developers.

"VirtualBox is a free Type 2 hypervisor by Oracle that lets you create and run virtual machines on your existing OS."

Xen

A Type 1 (bare-metal) hypervisor that runs directly on hardware. Used by Amazon AWS (old EC2 instances ran on Xen). Has a special "Dom0" — a privileged VM that controls all other VMs.

Why it is used: High performance, used in large data centers and cloud providers.

"Xen is a Type 1 bare-metal hypervisor that runs directly on hardware. It uses Dom0 as a privileged management VM to control all other guest VMs."

Hyper-V

Microsoft's Type 1 hypervisor built into Windows Server and Windows 10/11 Pro. You can enable it from Windows features and run VMs.

Why it is used: Already included in Windows, enterprise-grade, great for Windows-based VMs.

"Hyper-V is Microsoft's Type 1 bare-metal hypervisor built into Windows Server that allows you to run multiple virtual machines."

EC2 (Elastic Compute Cloud)

Amazon's service for renting virtual servers on the cloud. You choose how powerful a server you want, click a button, and in minutes you have a server running anywhere in the world.

Real-life example: Instead of buying a ₹50,000 server for your app, you rent an EC2 instance for ₹5/hour and delete it when done.

Why it is used: No upfront cost, launch in minutes, scale up/down anytime, worldwide availability.

"EC2 is Amazon's cloud service that provides resizable virtual servers (instances) on demand."

AMI (Amazon Machine Image)

A saved template/snapshot of a server — includes the OS, installed software, and configuration. You launch EC2 instances from AMIs.

Real-life example: Like a pre-installed disk image for your laptop. You use the same image to set up 100 identical laptops.

"AMI is a pre-configured template containing the OS and software used to launch EC2 instances."

EBS (Elastic Block Store)

A virtual hard drive that you attach to your EC2 instance. Stores your data persistently — even if you stop the instance, data remains.

Real-life example: Like a USB hard drive — you plug it into your computer (EC2), use it, unplug it, data stays.

Why it is used: Persistent storage for EC2, can be backed up using snapshots, can be resized.

"EBS is Amazon's block storage service — a virtual hard drive that attaches to EC2 instances and stores data persistently."

Security Groups

A virtual firewall that controls which traffic can reach your EC2 instance. You define rules — allow HTTP on port 80, allow SSH on port 22, etc.

Real-life example: Like a security guard at a building gate — only lets in people on the approved list (rules).

Why it is used: Protects your server from unauthorized access, controls inbound and outbound traffic.

"Security Groups are virtual firewalls that control inbound and outbound traffic to EC2 instances using rules."

IAM (Identity and Access Management)

AWS's system for controlling WHO can do WHAT in your AWS account. You create users, put them in groups, and attach permissions (policies) to groups.

Real-life example: Like an office access card system — HR has access to HR software, developers have access to servers, accountants have access to finance — all controlled centrally.

Why it is used: Security — ensures each person/service only has the minimum permissions they need.

"IAM is AWS's service for managing user identities and their permissions to access AWS resources."

VPC (Virtual Private Cloud)

Your own private, isolated section of the AWS cloud. Like having your own private data center inside AWS — with your own network, subnets, firewalls.

Real-life example: Like renting a private office floor in a big building — you have your own space, your own locked doors, your own network.

Why it is used: Network isolation, security, control over IP addressing and routing.

"VPC is a logically isolated virtual network within AWS where you can launch and manage your resources securely."

Public vs Private Subnet

Simple explanation:

- **Public Subnet:** Connected to the internet via an Internet Gateway. Servers here have public IPs. Web servers go here.
- **Private Subnet:** NOT directly connected to internet. Databases go here — safe from direct internet access.

Real-life example: Public subnet = reception area of an office (anyone can walk in). Private subnet = server room (only authorized people).

"Public subnets have internet access via an Internet Gateway; private subnets are isolated from direct internet access and used for backend/database servers."

Load Balancer

Distributes incoming traffic evenly across multiple servers so no single server gets overloaded.

Real-life example: Like a traffic policeman at a busy intersection — directs cars to different roads so no one road gets jammed.

Why it is used: High availability, prevents server overload, improves performance.

"A load balancer distributes incoming network traffic evenly across multiple servers to ensure no single server is overwhelmed."

Auto Scaling

Automatically adds more servers when traffic increases and removes them when traffic decreases. No manual work.

Real-life example: Like a restaurant that hires extra waiters during lunch rush and sends them home after — automatically.

Why it is used: Handles traffic spikes automatically, saves cost during low traffic, ensures availability.

"Auto Scaling automatically adjusts the number of EC2 instances based on demand to maintain performance and minimize cost."

CloudWatch

AWS's monitoring service. Watches your EC2 instances, collects metrics (CPU usage, network, disk), creates graphs, and triggers alarms when something goes wrong.

Real-life example: Like a doctor monitoring a patient's heart rate — if it goes above 70%, the alarm rings.

Why it is used: Monitor performance, detect problems early, trigger automatic actions via alarms.

"CloudWatch is AWS's monitoring service that collects metrics, creates dashboards, and triggers alarms based on defined thresholds."

S3 (Simple Storage Service)

AWS's online storage service for storing any type of file — images, videos, documents, backups. Files are stored as "objects" in "buckets."

Real-life example: Like Google Drive — but for your applications and AWS services.

Why it is used: Unlimited storage, 11 nines durability, cheap, used for static websites, backups, big data.

"S3 is Amazon's object storage service that stores files as objects in buckets with 99.999999999% durability."

IaaS, PaaS, SaaS

Model	You manage	AWS Example	Real-life analogy
-------	------------	-------------	-------------------

IaaS	OS, apps, data	EC2, EBS, VPC	Renting a flat — you furnish and manage it
PaaS	Only your app/code	Elastic Beanstalk, RDS	Renting a furnished flat — just move in
SaaS	Nothing — just use it	Gmail, Office 365	Staying in a hotel — everything provided

- *"IaaS provides virtualized hardware — you manage OS upward."*
- *"PaaS provides a platform — you just deploy your code."*
- *"SaaS provides ready-to-use software over the internet."*

Storage as a Service

Cloud storage provided on demand — you don't buy hard drives, you just use storage and pay per GB. AWS S3 and EBS are examples.

"Storage as a Service is cloud-based storage available on demand without managing physical hardware — S3 and EBS are AWS examples."

Database as a Service (DBaaS)

A managed database in the cloud. AWS handles installation, patching, backups, scaling — you just connect and run queries.

Why it is used: No DBA needed, automatic backups, high availability, easy to scale.

"DBaaS is a managed cloud database service where the provider handles maintenance — AWS RDS is an example."

SQL vs NoSQL

Feature	SQL	NoSQL
Structure	Tables with rows/columns	Documents, key-value, etc.
Schema	Fixed	Flexible
Example	MySQL, RDS	DynamoDB, MongoDB
Best for	Structured data, transactions	Unstructured, high-speed data
Scales	Vertically (bigger server)	Horizontally (more servers)

"SQL databases use structured tables with fixed schema; NoSQL databases use flexible formats like documents or key-value pairs and scale horizontally."

Containerization

Packaging your application and all its dependencies together in a small, portable "container" that runs the same everywhere.

Real-life example: Like a tiffin box — your food (app) and all accompaniments (dependencies) are packed together. Open it anywhere — same meal.

Why it is used: Solves "works on my machine" problem. Lightweight, fast, portable.

"Containerization packages an application with all its dependencies so it runs consistently across any environment."

Docker

The most popular tool for creating and running containers. You write a Dockerfile (recipe), Docker builds an image (packed box), you run the image as a container (open box).

Real-life example: Dockerfile = recipe. Docker image = frozen meal. Docker container = heated and served meal.

"Docker is a platform for building, shipping, and running applications in lightweight, portable containers."

Kubernetes

A system that automatically manages many containers running across many machines. Makes sure containers are always running, scales them, restarts failed ones.

Real-life example: If Docker is managing one chef, Kubernetes manages the entire restaurant chain — 100 chefs across 50 locations, all coordinated automatically.

Why it is used: Managing containers at scale, auto-healing, auto-scaling, load balancing.

"Kubernetes is an open-source container orchestration system that automates deployment, scaling, and management of containerized applications."

Pods

The smallest unit in Kubernetes. A Pod is a wrapper around one or more containers that share the same network and storage.

Real-life example: A Pod is like a room in a hostel — one or more students (containers) share the same room (network/storage).

"A Pod is the smallest deployable unit in Kubernetes — it contains one or more containers that share network and storage."

Deployment (Kubernetes)

A Kubernetes object that ensures a specific number of Pods are always running. If one Pod dies, Deployment automatically creates a new one.

"A Kubernetes Deployment manages Pod replicas, ensures the desired number are always running, and handles rolling updates."

Authentication vs Authorization

Simple explanation:

- **Authentication** = Proving WHO you are (login with username/password/MFA)
- **Authorization** = Deciding WHAT you can do after you're logged in (read-only vs admin)

Real-life example: Authentication = showing your ID card at office gate. Authorization = which floors/rooms you can enter.

"Authentication verifies identity — who you are. Authorization defines permissions — what you can do."

Scalability vs Elasticity

Feature	Scalability	Elasticity
Meaning	Ability to handle growing load	Automatically scale up AND down
Manual?	Can be manual	Automatic
Example	Adding more EC2 instances	Auto Scaling Group

"Scalability is the ability to handle increased load by adding resources. Elasticity is the automatic scaling up and down based on real-time demand."

STEP 2 — OVERALL IMPORTANT VIVA QUESTIONS

Q: What is cloud computing?

- **Easy** It means using computer services like servers, storage, and software over the internet instead of buying and managing physical hardware.

- *"Cloud computing is on-demand delivery of IT resources over the internet with pay-as-you-go pricing."*

Q: What are the advantages of cloud computing?

- **Easy** No upfront cost, pay per use, access from anywhere, automatic scaling, no maintenance.
- *"Cost savings, scalability, high availability, no maintenance, global access."*

Q: What is the difference between IaaS, PaaS, and SaaS?

- *"IaaS gives virtual hardware — you manage OS and apps. PaaS gives a platform — you just deploy code. SaaS gives ready software — you just use it. EC2 is IaaS, Elastic Beanstalk is PaaS, Gmail is SaaS."*

Q: What is a hypervisor? Name its types.

- *"A hypervisor creates and manages virtual machines. Type 1 runs directly on hardware — examples: Xen, Hyper-V. Type 2 runs on host OS — examples: VirtualBox, KVM."*

Q: What is the difference between a VM and a container?

	VM	Container
Size	GBs	MBs
Startup	Minutes	Seconds
OS	Full guest OS	Shares host kernel
Isolation	Hardware level	Process level

- *"VMs run a full OS and are heavy. Containers share the host OS kernel, are lightweight, start in seconds."*

Q: What is IAM in AWS?

- *"IAM is Identity and Access Management — it controls who can access what in AWS. We create users, groups, roles and attach policies to control permissions."*

Q: What is MFA? Why is it important?

- *"MFA is Multi-Factor Authentication — it adds a second layer of security beyond just a password. Even if password is stolen, account stays safe."*

Q: What is a Security Group in AWS?

- *"A Security Group is a virtual firewall for EC2 instances. It defines rules for inbound and outbound traffic — like allowing HTTP on port 80 and SSH on port 22."*

Q: What is the difference between S3 and EBS?

	S3	EBS
Type	Object storage	Block storage
Access	Via internet/URL	Attached to EC2
Use case	Files, images, backups	Virtual hard drive for EC2

- *"S3 is object storage accessed via URL — for files, images, websites. EBS is block storage attached directly to EC2 — like a virtual hard drive."*

Q: What is an EBS Snapshot?

- *"An EBS snapshot is a backup of an EBS volume stored in S3. You can use it to restore data or create a new volume."*

Q: What is CloudTrail?

- *"CloudTrail is AWS's audit logging service. It records every API call made in your AWS account — who did it, when, from where."*

Q: What is the difference between CloudWatch and CloudTrail?

	CloudWatch	CloudTrail
Purpose	Monitoring performance	Auditing API calls
What it tracks	Metrics (CPU, memory)	Who did what (events)

- *"CloudWatch monitors performance metrics like CPU usage. CloudTrail audits user activity — records who made what API call."*

Q: What is Terraform?

- *"Terraform is an Infrastructure as Code tool by HashiCorp. You write configuration files to automatically provision AWS resources instead of clicking in the console."*

Q: What is Elastic Beanstalk?

- *"Elastic Beanstalk is AWS's PaaS service. You upload your application code and it automatically handles deployment, load balancing, scaling, and monitoring."*

STEP 3 — EXPERIMENT-WISE VIVA Q&A

Q1 — IAM: Users, Groups, Policies, MFA

Aim: To create IAM users and groups, attach permission policies, enforce MFA, and verify access control.

What happens: You create 3 groups with different permissions, create 3 users, assign each to a group, and add MFA enforcement so users can't do anything without MFA.

Q: What groups did you create and what permissions did each have?

- *"I created three groups. S3-Support with AmazonS3ReadOnlyAccess for user-1. EC2-Support with AmazonEC2ReadOnlyAccess for user-2. EC2-Admin with AmazonEC2FullAccess for user-3."*

Q: What is the MFA Enforcement Policy and how does it work?

- *"The MFA Enforcement Policy has two parts. First — it allows users to set up their own MFA device. Second — it denies every other action if the user has not signed in with MFA. This is done using the condition `aws:MultiFactorAuthPresent` set to false."*

Q: What is `aws:MultiFactorAuthPresent`?

- *"It is an IAM policy condition that checks whether the current session used MFA to authenticate. If it's false — meaning no MFA — the deny statement blocks all actions."*

Q: What is the difference between a managed policy and a custom policy?

- *"A managed policy is a pre-written policy provided by AWS — like AmazonS3ReadOnlyAccess. A custom policy is one you write yourself in JSON — like our MFA-Enforcement-Policy."*

Q: What happened when user-1 tried to access EC2?

- *"user-1 was denied because user-1 is in the S3-Support group which only has S3 read permission. Trying to access EC2 gave an 'Access Denied' error."*

Q: What is the principle of least privilege?

- *"Give each user or service only the minimum permissions needed to do their job — nothing more. This reduces security risk."*

Q: Why did we not attach permissions directly to users?

- *"Best practice is to attach permissions to groups, not individual users. This makes management easier — adding/removing a user from a group automatically grants/revokes permissions."*

Q: What is BoolIfExists in the MFA policy?

- *"BoolIfExists checks the condition even if the key doesn't exist in the session. This ensures even old sessions without MFA are denied."*

Q: What is the difference between authentication and authorization in IAM context?

- *"Authentication is the login process — verifying who you are using username, password, and MFA. Authorization is what IAM policies control — what actions you are allowed to perform after logging in."*

Scenario Q: A user is in two groups — one allows S3 access, one denies S3 access. What happens?

- *"The Deny wins. In IAM, an explicit Deny always overrides any Allow. The user cannot access S3."*

Q2 — EBS: Storage, Backup, Recovery

Aim: To provision an EBS volume, attach it to EC2, store data, create a snapshot backup, and recover data from the snapshot.

What happens: You create a virtual hard drive, attach it to your server, format it, write data, take a backup snapshot, then restore from that snapshot to prove recovery works.

Q: What is the first command you ran after attaching the EBS volume?

- *"lsblk — this lists all block devices attached to the instance so I can confirm the new volume is visible."*

Q: Why did you run mkfs before mounting?

- *"A new EBS volume is like a blank hard drive — it has no file system. sudo mkfs -t ext4 /dev/xvdf creates an ext4 file system on it so the OS can read and write files to it."*

Q: What does mount do?

- *"sudo mount /dev/xvdf /mnt/my-data makes the EBS volume accessible at the directory /mnt/my-data. Without mounting, you cannot read or write files to it."*

Q: What is an EBS Snapshot?

- *"A snapshot is a point-in-time backup of an EBS volume stored in S3. If data is lost or corrupted, you can create a new volume from the snapshot and recover the data."*

Q: Why did you NOT run mkfs on the restored volume?

- *"Because the restored volume already has a file system and data on it from the snapshot. Running mkfs again would format it and erase all the data — destroying the recovery."*

Q: What is the final verification step?

- *"cat /mnt/recovered-data/testfile.txt — if it shows 'This is my backup test', the data recovery was successful."*

Q: What is the difference between a snapshot and a backup?

- *"They serve the same purpose — protecting data. An EBS snapshot is incremental — it only stores the changes since the last snapshot, making it efficient. It's stored in S3 automatically."*

Q: What happens to EBS data if the EC2 instance is terminated?

- *"By default, the root EBS volume is deleted when the instance terminates. But additional EBS volumes — like the one we created — persist unless you manually delete them."*

Q: What does the Availability Zone matching mean?

- *"EBS volumes can only be attached to EC2 instances in the same Availability Zone. If the instance is in us-east-1a, the EBS volume must also be in us-east-1a."*

Scenario Q: Your EC2 instance crashed and data is lost. How do you recover?

- *"I go to Snapshots, select the latest snapshot, click Actions → Create volume from snapshot, ensure it's in the same AZ, attach it to a new EC2 instance, mount it without running mkfs, and verify the data."*

Q3 — S3: Store and Retrieve Files

Aim: To create an S3 bucket, upload files, configure public access, and verify file access via URL.

What happens: You create a bucket, upload images/documents, write a bucket policy to make files publicly readable, and verify by accessing the file URL in an incognito browser.

Q: What is an S3 bucket?

- *"A bucket is a container in S3 that holds objects (files). Bucket names must be globally unique across all AWS accounts worldwide."*

Q: Why did we uncheck 'Block all public access'?

- *"By default S3 blocks all public access for security. We unchecked it because the experiment requires files to be publicly accessible via URL. In production, you should only do this for intentionally public content."*

Q: What is the bucket policy we applied?

- *"We applied a policy with Effect Allow, Principal *, Action s3:GetObject on the bucket's objects. This allows anyone on the internet to read (GET) files from the bucket."*

Q: What does Principal: * mean in the bucket policy?

- *"It means everyone — any user, any service, any anonymous internet user can perform the allowed action."*

Q: What is the difference between a bucket policy and an ACL?

- *"A bucket policy is a JSON document attached to the bucket that defines access rules for the entire bucket or specific paths. ACL (Access Control List) is an older method for setting permissions on individual objects. AWS recommends bucket policies over ACLs now."*

Q: How did you verify the file is publicly accessible?

- *"I copied the Object URL from the S3 console and pasted it in an incognito browser window. If the file opened or downloaded without any login, public access is confirmed."*

Q: What is the S3 object URL format?

- *"It looks like: https://bucket-name.s3.amazonaws.com/filename. For example: https://ccl-practical-pratham-2026.s3.amazonaws.com/image.jpg"*

Scenario Q: The file URL gives an 'Access Denied' error even after applying the policy. Why?

- *"Two possible reasons — first, 'Block all public access' is still enabled at the bucket level, overriding the policy. Second, the bucket policy resource ARN is wrong — it must end with /* to apply to all objects inside the bucket."*

Q4 — RDS: MySQL Database

Aim: To create a managed MySQL database using Amazon RDS, connect to it, and run SQL queries.

What happens: You provision an RDS MySQL instance, configure security groups to allow database connections, connect via MySQL Workbench, create a table, insert records, and verify with SELECT.

Q: What is Amazon RDS?

- *"RDS is Amazon's Relational Database Service — a fully managed database service. AWS handles hardware, OS patching, backups, and failover. You just connect and run SQL queries."*

Q: What database engine did you use and why?

- *"We used MySQL because it is free, widely used, and the Free Tier template in RDS supports it easily for lab practicals."*

Q: What is db.t3.micro?

- *"It's the instance type for the RDS database — similar to EC2 instance types. db.t3.micro is the smallest, cheapest option and is within the AWS Free Tier limit."*

Q: Why did you set Public Access to Yes?

- *"We needed to connect to the database from our local MySQL Workbench outside of AWS. Public access allows an external IP to reach the RDS endpoint on port 3306."*

Q: What port does MySQL use?

- *"Port 3306 — this is the default MySQL port. We added an inbound rule in the security group to allow TCP traffic on port 3306 from our IP."*

Q: What SQL commands did you run?

- **Answer:**

```
CREATE DATABASE ccl_test;
USE ccl_test;
CREATE TABLE Students (
  id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(50),
  roll_no INT
);
INSERT INTO Students (name, roll_no) VALUES ('Pratham', 64);
SELECT * FROM Students;
```

Q: What is AUTO_INCREMENT?

- *"It automatically assigns the next number to the id column when a new record is inserted. So the first row gets id=1, second gets id=2, and so on — without manually specifying the id."*

Q: What is the difference between RDS and installing MySQL on EC2?

- *"With RDS, AWS manages everything — patching, backups, high availability, failover. With MySQL on EC2, you are responsible for all that yourself. RDS is easier but less flexible."*

Q: What is automatic backup in RDS?

- *"RDS automatically takes a daily snapshot of your database and stores it for the backup retention period. We set it to 1 day. This allows point-in-time recovery."*

Scenario Q: You can't connect to RDS from MySQL Workbench. What do you check?

- *"Three things — first, check if the RDS security group allows inbound on port 3306 from your IP. Second, check if Public Access is set to Yes. Third, verify the endpoint hostname is correct in Workbench."*

Q5 — S3 Static Website

Aim: To host a simple static website (HTML + image) on Amazon S3.

What happens: You create an S3 bucket, enable static website hosting, upload index.html and an image, apply a public read bucket policy, and access the website via the bucket endpoint URL.

Q: What is static website hosting in S3?

- *"S3 can serve HTML files directly as a website without any web server. You enable static website hosting on the bucket, set the index document as index.html, and S3 serves it at a public URL."*

Q: What is the website endpoint URL format?

- *"It looks like: http://bucket-name.s3-website-us-east-1.amazonaws.com. This is different from the S3 object URL — this one serves the HTML as a rendered webpage."*

Q: What is the index document and why is it important?

- *"The index document is the default page S3 serves when someone visits the website URL without specifying a file — like index.html. Without it, S3 doesn't know which file to show."*

Q: What is the difference between S3 object URL and S3 website endpoint URL?

- *"The object URL directly downloads/accesses a file. The website endpoint URL renders the HTML file as a webpage in the browser — it follows the index document setting and supports HTML content delivery."*

Q: Can S3 host dynamic websites?

- *"No. S3 static website hosting only serves static content — HTML, CSS, JavaScript, images. For dynamic content like PHP or Python, you need EC2 or Elastic Beanstalk."*

Scenario Q: The website URL shows the image but not the text. Why?

- *"Likely the image file name in the HTML code doesn't exactly match the uploaded file name — for example, my-image.jpg vs My-Image.JPG. S3 is case-sensitive."*

Q6 — CloudWatch: Monitoring and Alarms

Aim: To monitor EC2 metrics using CloudWatch, create a CPU alarm that triggers when CPU exceeds 70%, and verify it using a stress test.

What happens: You view EC2 metrics in CloudWatch, create an alarm for CPU > 70%, link it to an SNS topic that emails you, then artificially spike the CPU to trigger the alarm.

Q: What is CloudWatch?

- *"CloudWatch is AWS's monitoring service. It collects metrics from AWS resources like EC2, displays graphs, and lets you set alarms that trigger actions when thresholds are crossed."*

Q: What metrics did you monitor?

- *"CPUUtilization — percentage of CPU being used. DiskReadOps — number of disk read operations. NetworkIn and NetworkOut — volume of traffic coming in and going out of the instance."*

Q: What command did you use to spike the CPU?

- *"sha1sum /dev/zero & — this runs a compression calculation on infinite zeros in the background, using 100% of one CPU core immediately."*

Q: What does the & at the end of the command do?

- *"The & runs the command in the background. This lets you continue using the terminal while the CPU stress is happening."*

Q: What is SNS?

- *"SNS stands for Simple Notification Service. It sends notifications — we used it to send an email alert when the CPU alarm triggered. You subscribe your email to an SNS topic and confirm the subscription."*

Q: Why did you need to confirm the SNS subscription?

- *"AWS sends a confirmation email to prevent spam. If you don't click Confirm Subscription in that email, notifications won't be delivered."*

Q: What are the three states of a CloudWatch alarm?

- *"OK — metric is below threshold. In Alarm — metric crossed the threshold and alarm fired. Insufficient Data — not enough data points yet to evaluate the alarm."*

Q: How did you stop the CPU stress?

- *"killall sha1sum — this kills all running processes named sha1sum, stopping the CPU stress."*

Q: What is the alarm threshold you set?

- *"70% CPUUtilization. When CPU usage goes above 70%, the alarm state changes from OK to In Alarm and an email notification is sent via SNS."*

Scenario Q: The alarm didn't fire even though CPU was at 100%. Why?

- *"CloudWatch checks metrics every 5 minutes by default. The alarm may need 1-2 data points above the threshold before firing. Also, confirm the SNS email subscription was confirmed — unconfirmed subscriptions don't receive notifications."*

Q7 — EC2 Web Server + Instance Resize

Aim: To install Apache web server on EC2, host a webpage, then resize the instance type and EBS volume.

What happens: You connect to a running EC2 instance, install Apache (httpd), create a simple HTML page, open port 80 in the security group, verify the website in a browser, then stop the instance and change its type/EBS size.

Q: What commands did you run to set up Apache?

```
sudo yum update -y      # Update packages
sudo yum install -y httpd # Install Apache
sudo systemctl start httpd # Start Apache
sudo systemctl enable httpd # Auto-start on reboot
echo "<h1>Welcome</h1>" | sudo tee /var/www/html/index.html
```

Q: Where does Apache serve files from?

- *"/var/www/html/ is the default document root for Apache. Any HTML file placed here is served as a webpage when someone accesses the server's IP."*

Q: Why did you add port 80 to the security group?

- *"HTTP traffic runs on port 80. Without an inbound rule allowing TCP port 80, the security group blocks all web traffic and the browser cannot reach the Apache server."*

Q: How do you verify the web server is working?

- *"Copy the Public IPv4 address of the EC2 instance and paste it in a browser. If you see the 'Welcome' message, Apache is running and the security group is correctly configured."*

Q: Why must you STOP the instance before resizing?

- *"Changing the instance type requires stopping the instance — like turning off a computer before swapping its CPU. A running instance cannot have its instance type changed."*

Q: How did you resize the instance?

- *"Stop the instance → Actions → Instance Settings → Change instance type → select t3.micro → Apply → Start the instance again."*

Q: How did you resize the EBS volume?

- *"EC2 → Elastic Block Store → Volumes → select the volume → Actions → Modify volume → increase size from 8 GiB to 10 GiB → Modify → confirm."*

Q: Do you need to stop the instance to resize EBS?

- *"No — EBS volumes can be resized while the instance is running. But you need to stop the instance to change the instance type."*

Q: What is systemctl enable httpd?

- *"This makes Apache start automatically every time the instance reboots. Without this, Apache would stop after a reboot and the website would be down."*

Q8 — VPC: Custom Network with Subnets

Aim: To create a custom VPC with public and private subnets, NAT Gateway, Internet Gateway, security groups, and deploy a web server inside it.

What happens: You use the VPC Wizard to create a complete network — 2 public subnets, 2 private subnets, an Internet Gateway for public access, a NAT Gateway for private subnet internet access, then launch an EC2 web server into the public subnet using a User Data script.

Q: What is a VPC?

- *"VPC stands for Virtual Private Cloud. It's your own isolated private network within AWS where you control IP ranges, subnets, routing, and security."*

Q: What is an Internet Gateway?

- *"An Internet Gateway connects your VPC to the internet. Resources in public subnets route internet traffic through the Internet Gateway."*

Q: What is a NAT Gateway and why is it needed?

- *"NAT Gateway allows resources in private subnets to access the internet for outbound requests — like downloading software updates — without being directly reachable from the internet. It's placed in the public subnet."*

Q: What is the difference between public and private subnet?

- *"A public subnet has a route to the Internet Gateway — resources here can be accessed from the internet. A private subnet has no direct internet route — resources here are protected and only accessible internally."*

Q: What is a Route Table?

- *"A route table contains rules that determine where network traffic is directed. Public subnets have a route — 0.0.0.0/0 → Internet Gateway. Private subnets have — 0.0.0.0/0 → NAT Gateway."*

Q: What VPC did you create?

- *"CCL-Custom-VPC with 2 Availability Zones, 2 public subnets, 2 private subnets, and a NAT Gateway in 1 AZ."*

Q: What is User Data in EC2?

- *"User Data is a script that runs automatically when an EC2 instance launches for the first time. We used it to install and start Apache automatically without manually connecting — Infrastructure as Code concept."*

Q: What security group did you create?

- *"Web-Server-SG with inbound rules: HTTP on port 80 from 0.0.0.0/0 for web access, and SSH on port 22 from My IP for secure management access."*

Q: How did you verify the VPC web server works?

- *"After the instance launched, I copied the Public IPv4 DNS address and pasted it in the browser. It showed 'Success! Web Server launched in a Custom VPC' — confirming the Internet Gateway, route table, and security group are all correctly configured."*

Scenario Q: EC2 is in the VPC but website is not loading. What do you check?

- *"First — check the security group has HTTP port 80 open. Second — check the instance is in a public subnet. Third — check the subnet's route table has a route to the Internet Gateway. Fourth — check Auto-assign public IP was enabled when launching."*

Q9a — CloudTrail: Audit Logging

Aim: To set up AWS CloudTrail to log all API activity and verify logs in both Event History and S3.

What happens: You create a CloudTrail trail that records all management events across all regions, stores logs in an S3 bucket, perform an AWS action (stop/start EC2), then find that action logged in Event History and in S3 as .json.gz files.

Q: What is CloudTrail?

- *"CloudTrail is AWS's audit and compliance service. It records every API call — every action taken in your AWS account — who did it, what resource was affected, when, and from which IP."*

Q: What trail did you create?

- *"Pratham-Activity-Log — applied to all regions, storing logs in a new S3 bucket auto-named by AWS."*

Q: Why did you set 'Apply trail to all regions'?

- *"To ensure activity in any AWS region worldwide is captured. Without this, only the current region's activity is logged — someone could create resources in another region without being audited."*

Q: What action did you perform to test CloudTrail?

- *"I stopped and then started the EC2 instance. This generates StopInstances and StartInstances events that should appear in CloudTrail."*

Q: Where can you see CloudTrail logs?

- *"Two places — Event History in the CloudTrail console shows the last 90 days immediately. The S3 bucket contains permanent .json.gz log files organized by Region/Year/Month/Day."*

Q: What information does a CloudTrail log entry contain?

- *"The event name (like StopInstances), who performed it (username/role), when it happened (timestamp), the source IP address, and which resource was affected (instance ID)."*

Q: What is the difference between CloudTrail and CloudWatch?

- *"CloudWatch monitors performance — CPU, memory, disk. CloudTrail audits actions — who did what in AWS. CloudWatch answers 'how is my server performing?' CloudTrail answers 'who made that change?'"*

Q9b — Terraform: Infrastructure as Code

Aim: To use Terraform to automatically provision an EC2 instance with a security group using code instead of clicking in the console.

What happens: You install Terraform, write a main.tf configuration file defining an AWS provider, a security group, and an EC2 instance, then run init/plan/apply to automatically create those resources.

Q: What is Terraform?

- *"Terraform is an open-source Infrastructure as Code tool by HashiCorp. You describe your infrastructure in configuration files and Terraform automatically creates it on AWS."*

Q: What is Infrastructure as Code (IaC)?

- *"Instead of manually clicking in the AWS console to create resources, you write code (configuration files) that describes what you want. The tool then creates it automatically — reproducible, version-controlled, consistent."*

Q: What are the four Terraform commands you used?

- **Answer:**

terraform init → Downloads the AWS provider plugin

terraform plan → Shows what will be created/changed (preview)

terraform apply → Actually creates the resources on AWS

terraform destroy → Deletes all resources created by Terraform

Q: What does terraform init do?

- *"It initializes the working directory — downloads the required provider plugins (in our case, the AWS provider). Run this once before any other Terraform command."*

Q: What does terraform plan show?

- *"It shows a preview of what Terraform will create, change, or destroy — without actually making any changes. Like a dry run. Always run plan before apply to review changes."*

Q: What is a provider in Terraform?

- *"A provider is a plugin that lets Terraform talk to a specific cloud or service. We used the AWS provider — it knows how to create EC2, security groups, and other AWS resources."*

Q: What resources did your main.tf create?

- *"Three resources — first, an AWS provider configured for us-east-1. Second, a security group named allow_web_ssh with SSH (22) and HTTP (80) inbound rules. Third, an EC2 instance of type t2.micro using Amazon Linux 2 AMI."*

Q: Why is terraform destroy important?

- *"To avoid unexpected AWS charges. After the lab, running terraform destroy removes all resources created by Terraform in one command — cleaner and safer than deleting manually."*

Q: What is the difference between terraform apply and manually creating EC2?

- *"Manual creation requires clicking through many console screens each time. Terraform apply reads the configuration file and creates everything automatically — same configuration every time, no human error, can be version-controlled with Git."*

Scenario Q: Terraform apply fails with 'UnauthorizedOperation'. Why?

- *"The AWS credentials configured via aws configure don't have permission to create EC2 or security groups. The IAM user needs appropriate policies like AmazonEC2FullAccess attached."*

Q10 — Elastic Beanstalk: PaaS Deployment

Aim: To deploy a Python web application using AWS Elastic Beanstalk without managing servers.

What happens: You create an Elastic Beanstalk application, select Python platform, choose a sample application, and Elastic Beanstalk automatically provisions EC2, load balancer, security groups — you just access the domain URL.

Q: What is Elastic Beanstalk?

- *"Elastic Beanstalk is AWS's PaaS service. You upload your application code and it automatically handles infrastructure — EC2 provisioning, load balancing, auto-scaling, and monitoring. You focus only on code."*

Q: What did Elastic Beanstalk create automatically?

- *"It automatically created an EC2 instance, configured it with Python runtime, set up a security group, assigned a public IP, and deployed the sample application. All with no manual configuration."*

Q: What are the two IAM roles needed for Elastic Beanstalk?

- *"aws-elasticbeanstalk-service-role — allows Elastic Beanstalk to manage AWS resources on our behalf. aws-elasticbeanstalk-ec2-role — attached to the EC2 instances EB creates, allows them to call AWS services."*

Q: What does 'Environment health: Green' mean?

- *"Green means the environment is healthy — the application is deployed, the EC2 instance is running, and the load balancer health checks are passing. Yellow means degraded, Red means something is broken."*

Q: What is the difference between EC2 (IaaS) and Elastic Beanstalk (PaaS)?

- *"With EC2 I manually install OS, web server, runtime, deploy code, configure security groups — everything. With Elastic Beanstalk I just upload my code — it handles all the infrastructure automatically. EC2 gives more control, Beanstalk gives more convenience."*

Q: How did you access the deployed application?

- *"Elastic Beanstalk provides a domain URL like ccl-exam-app.eba-xxxx.us-east-1.elasticbeanstalk.com. Clicking it opens the running application in the browser."*

Q: Can you SSH into an Elastic Beanstalk EC2 instance?

- *"Yes — if you attached an EC2 key pair during setup. Elastic Beanstalk still creates regular EC2 instances behind the scenes. You can find them in the EC2 console and SSH in."*

Q: How do you clean up after the Elastic Beanstalk experiment?

- *"Actions → Terminate Environment. This deletes the EC2 instance, load balancer, and all resources EB created. Important to avoid charges."*

STEP 4 — IMPORTANT COMMANDS EXPLAINED

Command	What it does	Expected output
lsblk	Lists all block devices (disks) attached	Shows disk names like xvda (root), xvdf (new EBS)
sudo mkfs -t ext4 /dev/xvdf	Creates ext4 file system on new EBS volume	Formatting messages, inode creation
sudo mkdir /mnt/my-data	Creates a directory to mount the volume	No output (success)
sudo mount /dev/xvdf /mnt/my-data	Mounts EBS volume to directory	No output (success)
echo "text" sudo tee /file.txt	Writes text to a file	Prints the text
cat /mnt/recovered-data/testfile.txt	Reads file content	Shows "This is my backup test"
sudo yum update -y	Updates all packages on Amazon Linux	List of updated packages
sudo yum install -y httpd	Installs Apache web server	Installation confirmation

sudo systemctl start httpd	Starts Apache service	No output (success)
sudo systemctl enable httpd	Enables Apache to start on reboot	Creates symlink message
sudo tee /var/www/html/index.html	Writes HTML to Apache web root	Prints the HTML content
sha1sum /dev/zero &	Stresses CPU to 100% in background	Process ID printed
killall sha1sum	Stops the CPU stress process	Terminated message
top	Shows running processes and CPU usage	Live process list
terraform init	Downloads Terraform AWS provider	"Terraform has been successfully initialized"
terraform plan	Shows preview of resources to be created	List of resources with + signs
terraform apply	Creates AWS resources	Resource creation logs, "Apply complete"
terraform destroy	Deletes all Terraform-managed resources	Deletion logs, "Destroy complete"
terraform -v	Shows Terraform version	Version number
aws configure	Sets AWS credentials for CLI/Terraform	Prompts for Access Key, Secret, Region

STEP 5 — LAST MINUTE REVISION

Most Important Definitions

Term	Definition
Cloud Computing	On-demand IT services over internet, pay-as-you-go
Hypervisor	Software that creates and manages VMs
Type 1 Hypervisor	Runs directly on hardware — Xen, Hyper-V
Type 2 Hypervisor	Runs on host OS — VirtualBox, KVM
EC2	AWS virtual server service

AMI	Template to launch EC2 instances
EBS	Persistent block storage for EC2
EBS Snapshot	Point-in-time backup of EBS volume
S3	Object storage service, files stored in buckets
IAM	AWS identity and access management
MFA	Multi-factor authentication — extra security layer
Security Group	Virtual firewall for EC2
VPC	Isolated private network in AWS
Internet Gateway	Connects VPC to internet
NAT Gateway	Allows private subnet resources to access internet
CloudWatch	AWS monitoring — metrics and alarms
CloudTrail	AWS audit logging — records all API calls
SNS	Simple Notification Service — sends alerts/emails
RDS	Managed relational database service
Elastic Beanstalk	AWS PaaS — deploy code, it manages everything
Terraform	Infrastructure as Code tool by HashiCorp
IaC	Infrastructure as Code — define infra in code files
Container	Lightweight isolated app + dependencies package
Docker	Platform to build and run containers
Kubernetes	Container orchestration system
Pod	Smallest Kubernetes unit — wraps containers
IaaS	Cloud-provided hardware — you manage OS + apps
PaaS	Cloud-managed platform — you manage only code
SaaS	Ready-to-use software over internet

Most Important Differences

CloudWatch vs CloudTrail: CloudWatch = monitors PERFORMANCE (CPU, RAM). CloudTrail = monitors ACTIONS (who did what).

EBS vs S3: EBS = block storage, attached to EC2, like hard drive. S3 = object storage, accessed via URL, for files.

Stop vs Terminate EC2: Stop = like shutdown, data preserved, can restart. Terminate = permanent delete.

Public Subnet vs Private Subnet: Public = internet access via Internet Gateway. Private = no direct internet, uses NAT Gateway.

VM vs Container: VM = full OS, heavy, minutes to start. Container = shared kernel, lightweight, seconds to start.

Authentication vs Authorization: Authentication = who are you? Authorization = what can you do?

SQL vs NoSQL: SQL = fixed tables, ACID, vertical scaling. NoSQL = flexible, horizontal scaling, key-value.

IaaS vs PaaS: IaaS = you manage OS + runtime + app. PaaS = you manage only app code.

mkfs vs mount: mkfs = creates file system (one time on new disk). mount = makes disk accessible at a path.

terraform plan vs terraform apply: plan = preview only, no changes. apply = actually creates resources.

TOP 50 VIVA QUESTIONS — COMPLETE ANSWERS

1. What is cloud computing? Cloud computing is the delivery of computing services like servers, storage, databases, and software over the internet on a pay-as-you-go basis. Instead of buying physical hardware, you rent it from providers like AWS and pay only for what you use.

2. What are the 5 characteristics of cloud computing?

- **On-demand self-service** — you provision resources yourself without calling anyone
- **Broad network access** — accessible from anywhere via internet
- **Resource pooling** — one physical infrastructure serves many customers
- **Rapid elasticity** — scale up or down quickly based on need
- **Measured service** — you pay only for what you use, like a utility bill

3. Difference between IaaS, PaaS, SaaS?

- **IaaS** — you get virtual hardware. You manage OS, runtime, and apps. Example: EC2.
- **PaaS** — you get a platform. You only manage your code. Example: Elastic Beanstalk.
- **SaaS** — you get ready-to-use software. You manage nothing. Example: Gmail.

4. What is virtualization? Virtualization is the process of running multiple virtual machines on a single physical machine using software called a hypervisor. It divides one physical machine's resources — CPU, RAM, storage — into multiple isolated virtual machines.

5. What is a hypervisor? A hypervisor is software that creates and manages virtual machines. It sits between the physical hardware and the virtual machines, allocating resources like CPU and RAM to each VM.

6. Difference between Type 1 and Type 2 hypervisor?

- **Type 1** runs directly on bare hardware — no host OS needed. It's faster and used in data centers. Examples: Xen, Hyper-V.
- **Type 2** runs on top of an existing host OS. It's slower but easier to set up on personal computers. Examples: VirtualBox, KVM.

7. Examples of Type 1 and Type 2 hypervisors?

- Type 1: Xen, Hyper-V, VMware ESXi
- Type 2: VirtualBox, KVM, VMware Workstation

8. What is a Virtual Machine? A virtual machine is a software-based computer that runs its own operating system inside a physical computer. It behaves exactly like a real computer but shares the underlying hardware with other VMs through a hypervisor.

9. What is Host OS vs Guest OS?

- **Host OS** is the operating system running on the actual physical hardware — for example, Windows on your laptop.
- **Guest OS** is the operating system running inside a virtual machine — for example, Ubuntu running inside VirtualBox on that same laptop.

10. What is KVM? KVM stands for Kernel-based Virtual Machine. It is a Type 2 hypervisor built directly into the Linux kernel. When KVM is enabled, Linux itself becomes capable of running virtual machines with near-hardware performance by using the CPU's built-in virtualization features like Intel VT-x.

11. What is Xen? What is Dom0? Xen is a Type 1 bare-metal hypervisor that runs directly on hardware. Dom0 — Domain Zero — is a special privileged virtual machine in Xen that has direct access to hardware and is responsible for managing all other guest VMs, called DomU. Without Dom0, Xen cannot function.

12. What is EC2? EC2 stands for Elastic Compute Cloud. It is Amazon's service for renting virtual servers in the cloud. You choose the size, OS, and region, launch an instance in minutes, and pay only for the time it runs.

13. What is an AMI? AMI stands for Amazon Machine Image. It is a pre-configured template that contains the operating system and software needed to launch an EC2 instance. You select an AMI when launching EC2 — for example, Amazon Linux 2023 AMI.

14. What is EBS? EBS stands for Elastic Block Store. It is Amazon's block storage service — a virtual hard drive that you attach to an EC2 instance. Data on EBS persists even when the instance is stopped or restarted.

15. What is an EBS Snapshot? An EBS snapshot is a point-in-time backup of an EBS volume. It is stored in S3 automatically. You can use a snapshot to restore data by creating a new EBS volume from it, which is exactly what we did in Experiment 2.

16. Why not run mkfs on a restored volume? Because the restored volume already has a filesystem and all the original data on it from the snapshot. Running mkfs would format the disk and permanently erase all that data — destroying the entire purpose of the recovery. You just mount it directly without formatting.

17. What is S3? S3 stands for Simple Storage Service. It is Amazon's object storage service where you store files — called objects — inside containers called buckets. It offers 99.999999999% durability and is used for backups, static websites, and file storage.

18. What is a bucket policy? What does Principal:* mean? A bucket policy is a JSON document attached to an S3 bucket that defines who can access it and what actions they can perform. Principal:* means everyone — any user, any AWS account, or any anonymous internet user. In our experiment, we used it to allow public read access to files in the bucket.

19. What is IAM? IAM stands for Identity and Access Management. It is AWS's service for controlling who can access what in your AWS account. You create users, organize them into groups, and attach permission policies to control their access to AWS resources.

20. What are IAM users, groups, roles, policies?

- **User** — an individual person or application with login credentials
- **Group** — a collection of users that share the same permissions
- **Role** — an identity with temporary permissions, assumed by AWS services or applications
- **Policy** — a JSON document that defines what actions are allowed or denied on which resources

21. What is MFA and why is it needed? MFA stands for Multi-Factor Authentication. It adds a second layer of security beyond just a password — you need both your password and a code from an authenticator app. It is needed because if a password is stolen or guessed, the attacker still cannot log in without the second factor.

22. What is the MFA enforcement policy condition? The condition is `aws:MultiFactorAuthPresent` set to false inside a Deny statement. This means — if the user logged in WITHOUT MFA, deny all actions. We used `BoolIfExists` so it also applies to sessions where the MFA key doesn't even exist, covering all cases.

23. Explicit Deny vs Allow — what wins? Explicit Deny always wins. In AWS IAM, if there is even one Deny statement that applies to an action, it overrides every Allow statement. So in our experiment, when user-1 in S3-Support tried to access EC2, the implicit deny blocked it — and with the MFA policy, an explicit deny blocked everything for users without MFA.

24. What is a Security Group? A Security Group is a virtual firewall that controls inbound and outbound traffic for EC2 instances. You define rules — for example, allow HTTP on port 80 from anywhere, allow SSH on port 22 from your IP only. Security Groups are stateful — if you allow inbound traffic, the response is automatically allowed outbound.

25. Difference between Security Group and NACL?

- **Security Group** works at the instance level, is stateful — responses are automatically allowed
- **NACL** (Network Access Control List) works at the subnet level, is stateless — you must define both inbound and outbound rules explicitly
- Security Groups only allow rules; NACLs can explicitly deny

26. What is a VPC? VPC stands for Virtual Private Cloud. It is a logically isolated private network within AWS that you fully control. You define your own IP address range, create subnets, set up routing, and configure security. In Experiment 8 we created CCL-Custom-VPC.

27. What is an Internet Gateway? An Internet Gateway is a VPC component that allows resources in your public subnets to communicate with the internet. It provides two-way communication — traffic can come in from the internet and go out to the internet through it.

28. What is a NAT Gateway? A NAT Gateway allows resources in private subnets to make outbound connections to the internet — like downloading software updates — but prevents the internet from initiating inbound connections to those resources. It is placed in a public subnet and provides one-way outbound internet access for private resources.

29. Difference between public and private subnet? A public subnet has a route to the Internet Gateway in its route table, so resources here can be reached from the internet and can reach the internet. A private subnet has no direct internet route — resources here are protected and only accessible within the VPC. Web servers go in public subnets; databases go in private subnets.

30. What is a route table? A route table contains rules that tell network traffic where to go. For public subnets the route is: all internet traffic (0.0.0.0/0) goes to the Internet Gateway. For private subnets the route is: all internet traffic goes to the NAT Gateway. Every subnet must be associated with a route table.

31. What is CloudWatch? CloudWatch is AWS's monitoring and observability service. It collects performance metrics from AWS resources like EC2, displays them as graphs, and lets you create alarms

that trigger notifications or actions when a metric crosses a defined threshold. In Experiment 6 we used it to monitor CPU usage.

32. What metrics did you monitor in CloudWatch? We monitored three metrics from our EC2 instance:

- **CPUUtilization** — percentage of CPU being used
- **DiskReadOps** — number of read operations from EBS
- **NetworkIn / NetworkOut** — volume of traffic flowing into and out of the instance

33. What command spikes CPU? How do you stop it? `sha1sum /dev/zero &` — this calculates a hash on infinite zeros, consuming 100% of one CPU core. The `&` runs it in the background. To stop it, run `killall sha1sum` which terminates all running `sha1sum` processes.

34. What is SNS? Why confirm subscription? SNS stands for Simple Notification Service. It delivers notifications to subscribers — in our case, email alerts when the CloudWatch alarm fires. You must confirm the subscription by clicking the link in the confirmation email AWS sends. Without confirmation, the email endpoint is not active and notifications will not be delivered.

35. What are CloudWatch alarm states? There are three states:

- **OK** — the metric is within the normal threshold
- **In Alarm** — the metric has crossed the threshold and the alarm has fired
- **Insufficient Data** — there is not enough data yet to evaluate the alarm condition

36. What is CloudTrail? CloudTrail is AWS's audit and governance service. It records every API call made in your AWS account — who performed it, what action was taken, which resource was affected, when it happened, and from which IP address. In Experiment 9a we created a trail named Pratham-Activity-Log.

37. Difference between CloudWatch and CloudTrail? CloudWatch monitors **performance** — it tracks metrics like CPU usage, memory, and disk I/O and answers "how is my resource performing?" CloudTrail monitors **activity** — it logs every action taken in the account and answers "who did what and when?" CloudWatch is for operations; CloudTrail is for security and compliance auditing.

38. Where are CloudTrail logs stored? CloudTrail logs are stored in two places. First, Event History in the CloudTrail console — shows the last 90 days of activity immediately and is searchable. Second, an S3 bucket — permanent storage where logs are delivered as compressed `.json.gz` files organized in folders by Region, Year, Month, and Day.

39. What is Terraform? Terraform is an open-source Infrastructure as Code tool made by HashiCorp. Instead of clicking through the AWS console to create resources, you write configuration files describing what you want, and Terraform automatically creates those resources on AWS. In Experiment 9b we wrote a `main.tf` file that created a security group and an EC2 instance.

40. What is Infrastructure as Code? Infrastructure as Code means defining and managing your cloud infrastructure using code files instead of manual steps. The configuration is version-controlled, repeatable, and consistent — you can recreate the exact same infrastructure every time by running the same code. It eliminates human error from manual configuration.

41. What do terraform init, plan, apply, destroy do?

- **terraform init** — initializes the project and downloads the required provider plugins (like the AWS provider)
- **terraform plan** — shows a preview of what will be created, changed, or deleted — no actual changes happen
- **terraform apply** — executes the plan and actually creates the resources on AWS
- **terraform destroy** — deletes all resources that were created by Terraform

42. What is Elastic Beanstalk? Elastic Beanstalk is AWS's Platform as a Service. You upload your application code — Python, Node.js, PHP, etc. — and Elastic Beanstalk automatically handles everything else: provisioning EC2 instances, configuring the web server, setting up load balancing, auto-scaling, and health monitoring. In Experiment 10 we deployed a Python sample application using it.

43. What IAM roles does Elastic Beanstalk need? Two roles are needed:

- **aws-elasticbeanstalk-service-role** — allows Elastic Beanstalk itself to manage AWS resources like EC2 and load balancers on your behalf
- **aws-elasticbeanstalk-ec2-role** — attached to the EC2 instances that Elastic Beanstalk creates, allowing them to interact with other AWS services

44. What does 'Environment health: Green' mean? Green means the environment is completely healthy — the EC2 instance is running, the application is deployed successfully, and all health checks are passing. Yellow means the environment is degraded with some issues. Red means it is severely broken. Grey means the environment is updating or the health is unknown.

45. Difference between EC2 and Elastic Beanstalk? With EC2 you manage everything yourself — you install the OS, web server, runtime, deploy code, configure security groups, and handle scaling manually. With Elastic Beanstalk you only upload your code — it automatically handles the EC2 instances, web server, load balancer, scaling, and monitoring. EC2 gives full control, Elastic Beanstalk gives convenience and speed.

46. What is Docker? Docker is a platform for building, shipping, and running applications in containers. You write a Dockerfile that describes your application and its dependencies, Docker builds it into an image, and you run that image as a container anywhere — on your laptop, on a server, or in the cloud — and it behaves exactly the same.

47. What is a container? A container is a lightweight, portable package that includes an application and all its dependencies — libraries, configuration, runtime — everything needed to run. Containers share the host machine's OS kernel, making them much smaller and faster than virtual machines.

48. Difference between VM and container?

- A VM contains a full guest OS — it is large (GBs), takes minutes to start, and has strong hardware-level isolation
- A container shares the host OS kernel — it is small (MBs), starts in seconds, and uses process-level isolation
- VMs are heavier and more isolated; containers are lightweight and more portable

49. What is Kubernetes? Kubernetes is an open-source container orchestration system. When you have many containers running across many machines, Kubernetes manages them automatically — it ensures the right number of containers are always running, restarts failed ones, distributes traffic, and scales containers up or down based on demand.

50. What is authentication vs authorization? Authentication is the process of verifying who you are — for example, logging in with a username, password, and MFA code. Authorization is the process of deciding what you are allowed to do after you are authenticated — for example, IAM policies that say user-1 can only read S3 but cannot touch EC2. Authentication comes first, authorization comes after.

FAST REVISION — ONE-LINERS

- Cloud = renting IT over internet
- Hypervisor = software that runs VMs
- Type 1 = bare metal (Xen, Hyper-V), Type 2 = on host OS (VirtualBox, KVM)
- EC2 = AWS virtual server
- AMI = launch template for EC2
- EBS = virtual hard drive for EC2
- Snapshot = EBS backup stored in S3
- Never mkfs a restored volume — it erases data
- S3 = object storage in buckets, globally unique names
- Principal:* = allow everyone
- IAM = who can do what in AWS
- Explicit Deny always beats Allow
- MFA = password + physical device = more secure
- Security Group = stateful firewall for EC2
- VPC = your private network in AWS
- Internet Gateway = connects VPC to internet
- NAT Gateway = lets private subnets reach internet
- Public subnet = has internet route, Private = doesn't
- CloudWatch = monitors performance metrics
- `sha1sum /dev/zero &` = spikes CPU, `killall sha1sum` = stops it
- SNS = sends email notifications for alarms
- CloudTrail = audit log of all AWS API calls
- Terraform = write code to create AWS infra
- `terraform init` → `plan` → `apply` → `destroy`

- Elastic Beanstalk = upload code, AWS manages everything
- Container = lightweight app + dependencies package
- VM = full OS, Container = shared kernel, much lighter

STEP 6 — FINAL ANALYSIS

HIDDEN TOPICS PROFESSORS COMMONLY ASK

About EBS

Q: What is the difference between gp2 and gp3 EBS volumes?

Both are SSD-based general purpose volumes but gp3 is the newer, better, and cheaper version.

- **gp2** — performance is tied to size. Bigger the volume, more IOPS you get automatically. You cannot set them separately.
- **gp3** — you can set IOPS and throughput independently of size. Cheaper than gp2 and gives better baseline performance.

"gp2 links performance to volume size — bigger size means more IOPS automatically. gp3 lets you configure IOPS and throughput independently of size and is cheaper. gp3 is the current recommended default."

Q: Can one EBS volume be attached to two EC2 instances at the same time?

Normally NO — one EBS volume can only be attached to one EC2 instance at a time. But there is a special exception — io2 Block Express volumes support a feature called **Multi-Attach** which lets one volume attach to up to 16 instances in the same Availability Zone. But this is advanced and requires the application to handle concurrent writes carefully.

"Generally no — one EBS volume attaches to one EC2 instance at a time. The exception is io2 volumes with Multi-Attach enabled, which allows up to 16 instances in the same AZ to share one volume."

Q: What happens to an EBS volume when you terminate an EC2 instance?

It depends on a setting called **Delete on Termination**.

- For the **root volume** — Delete on Termination is ON by default, so it gets deleted when the instance terminates.
- For **additional volumes** we attach — Delete on Termination is OFF by default, so they survive and remain available.

"The root EBS volume is deleted by default when an instance terminates. Additional EBS volumes we attach separately persist unless we manually set Delete on Termination to true."

Q: What is EBS encryption?

You can encrypt an EBS volume so all data stored on it is encrypted at rest. AWS uses KMS keys to manage encryption. If someone physically steals the disk, they cannot read the data.

"EBS encryption encrypts data at rest on the volume using AWS KMS keys. It protects data from unauthorized physical access and is transparent to the instance — no changes needed in the application."

About S3

Q: What is S3 versioning?

Versioning keeps every version of every file you upload. If you accidentally delete or overwrite a file, you can go back and restore the previous version. Like an undo button for your files.

"S3 versioning keeps multiple versions of the same object. If a file is accidentally deleted or overwritten, you can restore any previous version. Once enabled, it cannot be disabled — only suspended."

Q: What are S3 storage classes?

S3 has different storage tiers with different costs and retrieval speeds. You pick based on how often you access the data.

Storage Class	Use case	Retrieval
Standard	Frequently accessed	Instant
Standard-IA	Infrequently accessed	Instant but cheaper
Glacier	Archives, rarely accessed	Minutes to hours
Glacier Deep Archive	Long-term archive	Up to 12 hours
Intelligent-Tiering	Unknown access pattern	Auto-moves between tiers

"S3 has multiple storage classes — Standard for frequent access, Standard-IA for infrequent, Glacier for archival. They differ in cost and retrieval time. You choose based on how often you need the data."

Q: What is the maximum size of a single S3 object?

"The maximum size of a single S3 object is 5 terabytes. For objects larger than 5 GB, you must use multipart upload which splits the file into parts and uploads them in parallel."

Q: What is S3 durability vs availability?

Easy answer:

- **Durability** = will your data get lost? S3 durability is 99.999999999% — 11 nines. Meaning AWS stores copies across multiple data centers so data is almost never lost.
- **Availability** = can you access it right now? Standard S3 availability is 99.99% — meaning it's accessible almost all the time.

"S3 durability is 99.999999999% — 11 nines — meaning data is almost never lost because AWS stores multiple copies. Availability is 99.99% meaning the service is accessible nearly all the time."

About VPC

Q: What is the CIDR block of your VPC?

CIDR block defines the IP address range of your VPC. The default VPC uses **172.31.0.0/16** — meaning it can have IP addresses from 172.31.0.0 to 172.31.255.255. The /16 means the first 16 bits are fixed and the rest are available for subnets and instances.

"The default VPC CIDR is 172.31.0.0/16. In our custom VPC experiment we created CCL-Custom-VPC. The /16 means 65,536 possible IP addresses within that network."

Q: What is the difference between Internet Gateway and NAT Gateway?

Easy answer:

- **Internet Gateway** — works two ways. Traffic can come IN from internet and go OUT to internet. Used by public subnets.
- **NAT Gateway** — works one way. Traffic can only go OUT to internet. Internet cannot initiate connections IN. Used by private subnets for updates/downloads without exposing them.

"Internet Gateway provides bidirectional internet access for public subnets. NAT Gateway provides outbound-only internet access for private subnets — resources can reach internet but internet cannot reach them directly."

Q: What is a NACL and how is it different from a Security Group?

Easy answer:

- **Security Group** — works at instance level, stateful (return traffic automatically allowed), only allows rules — no explicit deny
- **NACL** — works at subnet level, stateless (you must allow both inbound and outbound separately), supports both allow and deny rules

"Security Groups are stateful instance-level firewalls — return traffic is automatically allowed. NACLs are stateless subnet-level firewalls — you must explicitly allow both inbound and outbound, and they support deny rules."

● About Terraform

Q: What is a .tfstate file?

When Terraform creates resources, it saves a record of everything it created in a file called **terraform.tfstate**. This is Terraform's memory — it knows what exists, what needs to be updated, and what needs to be deleted. Without this file, Terraform doesn't know what it previously created.

"terraform.tfstate is Terraform's state file. It records all resources Terraform has created. Terraform uses it to compare the current state with the desired state and decide what changes to make. Never delete it manually."

Q: What is the difference between terraform plan and terraform apply?

"terraform plan is a dry run — it shows what Terraform will create, change, or destroy without actually doing anything. terraform apply actually executes those changes and creates the real resources on AWS. Always run plan before apply to review changes."

Q: What is a resource block in Terraform?

A resource block in Terraform defines one AWS resource you want to create. For example, resource "aws_instance" "my_ec2" tells Terraform to create an EC2 instance. The first part is the resource type, the second is the local name you give it.

"A resource block in Terraform defines an infrastructure component to create. It has a type like aws_instance and a local name. Inside the block you specify configuration like AMI, instance type, and security groups."

● About Elastic Beanstalk

Q: What happens behind the scenes when you deploy on Elastic Beanstalk?

When you submit the application, Elastic Beanstalk automatically creates several AWS resources on your behalf — an EC2 instance to run the app, a security group to control traffic, an Auto Scaling group to manage scaling, a load balancer to distribute traffic, and a CloudWatch alarm to monitor health. All of this happens automatically from one upload.

"Behind the scenes Elastic Beanstalk creates an EC2 instance, security group, Auto Scaling group, load balancer, and CloudWatch alarms automatically. You only upload code — it manages all the underlying infrastructure."

Q: What is the difference between Elastic Beanstalk and EC2?

"EC2 is IaaS — you manage everything from OS to deployment manually. Elastic Beanstalk is PaaS — you only upload your application code and it automatically handles EC2, web server, scaling, and monitoring. EC2 gives more control, Beanstalk gives more convenience."

Q: Can you customize the EC2 instance inside Elastic Beanstalk?

"Yes — Elastic Beanstalk allows configuration of the underlying EC2 instance type, key pair, security groups, VPC, and scaling settings. You can also SSH into the instance if a key pair was configured during setup."

About IAM

Q: What is the root user? Should you use it daily?

The root user is the original AWS account owner login — it has unrestricted access to everything in the account and cannot be limited by any policy. You should never use it for daily work because if compromised, the attacker gets complete control. Create an IAM admin user for daily tasks and lock the root user with MFA.

"The root user has unrestricted access to everything in the AWS account. It should never be used for daily tasks — only for account-level tasks like billing. Always protect it with MFA and create IAM users for regular work."

Q: What is an IAM Role vs an IAM User?

Easy answer:

- **IAM User** — a permanent identity for a person. Has long-term credentials — username, password, access keys.
- **IAM Role** — a temporary identity meant to be assumed by AWS services, applications, or other accounts. Provides temporary credentials that expire. For example, an EC2 instance assumes a role to access S3 without hardcoding credentials.

"An IAM User is a permanent identity for a person with long-term credentials. An IAM Role is a temporary identity assumed by AWS services or applications — it provides short-term credentials and is more secure for service-to-service access."

Q: What is a policy document? What are its main elements?

"A policy document is a JSON file that defines permissions. Its main elements are: Version — the policy language version, Statement — a list of permission rules, Effect — Allow or Deny, Action — which AWS actions are controlled, Resource — which resources it applies to, and optionally Condition — when the rule applies."

About RDS

Q: What is Multi-AZ in RDS?

Multi-AZ means AWS automatically creates a standby copy of your database in a different Availability Zone. If the primary database fails — hardware issue, network problem — AWS automatically switches over to the standby with no data loss and minimal downtime. It is for high availability, not for performance.

"Multi-AZ in RDS maintains a standby database replica in a different Availability Zone. If the primary fails, RDS automatically fails over to the standby. It provides high availability and disaster recovery — not for read scaling."

Q: What is the difference between RDS and installing MySQL on EC2?

"With RDS, AWS manages patching, backups, Multi-AZ failover, and scaling. With MySQL on EC2, you manage everything yourself — installation, patching, backups, replication. RDS is easier and more reliable but less flexible and more expensive."

Q: What is a Read Replica in RDS?

A Read Replica is a copy of your database that handles only read queries — SELECT statements. It reduces load on the primary database. Unlike Multi-AZ which is for failover, Read Replicas are for performance and read scaling.

"A Read Replica is a read-only copy of your RDS database. It handles SELECT queries to reduce load on the primary. Unlike Multi-AZ which is for availability, Read Replicas are for read performance scaling."

QUESTIONS STUDENTS USUALLY FORGET

Q: Why is the Availability Zone important when creating an EBS volume?

EBS volumes can only be attached to EC2 instances in the SAME Availability Zone. If your EC2 is in us-east-1a and your EBS volume is in us-east-1b, you cannot attach it. This is why in Experiment 2 we made sure to select the same AZ as our running instance when creating the volume and when restoring from snapshot.

"EBS volumes are AZ-specific. A volume in us-east-1a can only be attached to an EC2 instance also in us-east-1a. This is why we carefully matched the AZ when creating the volume and when creating a volume from the snapshot."

Q: What is the SNS subscription confirmation for? What happens if you don't confirm?

When you create an SNS topic and add your email, AWS sends a confirmation email to verify you own that address and you actually want to receive notifications — to prevent spam. If you don't click Confirm Subscription in that email, your email is not activated as a subscriber and you will never receive alarm notifications even if the alarm fires.

"SNS sends a confirmation email to verify the subscriber actually wants notifications — preventing spam. If you don't confirm, the email endpoint stays pending and CloudWatch alarm notifications are never delivered to that email."

Q: What is 'Apply trail to all regions' in CloudTrail? Why enable it?

AWS has many regions worldwide. If you only apply CloudTrail to one region, actions taken in other regions — like someone quietly launching an instance in Tokyo — go unlogged. Applying to all regions means every action anywhere in the world is captured in one trail.

"Applying CloudTrail to all regions ensures API activity in every AWS region is captured. Without this, only the current region is audited — a malicious user could create resources in other regions without being logged."

Q: What is User Data in EC2? When does it run?

User Data is a script you provide when launching an EC2 instance. It runs automatically on the very first boot of the instance — only once. We used it in Experiment 8 to install Apache and create the webpage automatically without manually connecting to the instance.

"User Data is a bootstrap script that runs automatically on first launch of an EC2 instance. We used it in the VPC experiment to automatically install and start Apache — no manual SSH connection needed."

Q: Why did you run systemctl enable httpd? What happens without it?

systemctl start httpd only starts Apache right now for this session. If the instance reboots, Apache will not start automatically and the website goes down. systemctl enable httpd tells the OS to always start Apache automatically on every boot.

"systemctl start httpd starts Apache immediately. systemctl enable httpd makes it start automatically on every reboot. Without enable, the web server would be down every time the instance restarts."

Q: What is the difference between stopping and terminating an EC2 instance?

Easy answer:

- **Stop** — like shutting down your computer. The instance is off but not deleted. EBS data is preserved. You can start it again. You stop being charged for compute but still pay for EBS storage.
- **Terminate** — permanent deletion. The instance is gone forever. Root EBS volume is deleted by default. Cannot be undone.

"Stopping an EC2 instance is like shutting it down — data is preserved and you can restart it. Terminating is permanent deletion — the instance and its root volume are gone forever and cannot be recovered."

Q: What happens if you forget to open port 80 in the security group after installing Apache?

Apache is running perfectly on the server but the security group is blocking all traffic on port 80. When you try to access the public IP in a browser, it will just time out and show "This site can't be reached" — even though the server is fine. The fix is to add an inbound rule: HTTP, port 80, source 0.0.0.0/0.

"If port 80 is not open in the security group, the browser request never reaches Apache — it times out. Security Groups act as the first filter before traffic reaches the server. Adding HTTP port 80 inbound rule immediately fixes it."

Q: What is the bucket website endpoint URL vs the S3 object URL?

Easy answer:

- **Object URL** — direct link to a specific file: <https://bucket.s3.amazonaws.com/file.jpg> — downloads or shows that one file
- **Website endpoint URL** — serves the bucket as a website:
<http://bucket.s3-website-us-east-1.amazonaws.com> — loads index.html as a webpage, follows redirects, renders HTML properly

"The S3 object URL directly accesses a specific file. The website endpoint URL treats the bucket as a website — it serves index.html as the homepage and renders HTML. Static website hosting must be enabled for the endpoint to work."

Q: In Terraform, what does the AMI ID represent and why does it matter?

The AMI ID is the unique identifier for the machine image used to launch your EC2 instance. In our Terraform file we used `ami-0c55b159cbfafa1f0` which is the Amazon Linux 2 AMI for us-east-1. AMI IDs are region-specific — the same AMI has a different ID in different regions. Using the wrong AMI ID causes the apply to fail.

"The AMI ID uniquely identifies the machine image used to launch the EC2 instance. It is region-specific — the same OS image has a different ID in each region. Using an AMI ID from the wrong region will cause terraform apply to fail."

Q: What is the Default VPC? Why did we use it in Elastic Beanstalk?

AWS automatically creates one Default VPC in every region when you create an account. It has default subnets in every Availability Zone and an Internet Gateway already configured. We used it in Elastic Beanstalk because it is the simplest option for a lab — everything is pre-configured, no setup needed.

"The Default VPC is automatically created by AWS in every region. It comes pre-configured with subnets in every AZ and an Internet Gateway. We used it for Elastic Beanstalk to keep the lab simple — no custom networking setup needed."

Q: Why did we select 'Free Tier' template in RDS?

The Free Tier template automatically disables expensive options like Multi-AZ deployment and storage autoscaling, and limits the instance to db.t3.micro — the smallest free option. Without selecting Free Tier, you might accidentally enable options that cost money.

"Selecting Free Tier in RDS automatically restricts the configuration to free options — db.t3.micro instance, 20 GiB storage, single AZ, no expensive add-ons. It prevents accidental charges during the lab."

Q: What is the difference between terraform init and terraform plan?

"terraform init sets up the working directory — downloads the provider plugins needed to talk to AWS. It runs once. terraform plan reads your configuration and shows a preview of what will be created or changed — no resources are actually touched. Always init first, then plan, then apply."